

IM_IMFI - Backup Verification Simulator (IM06)– Project Description

Project Overview

Backup Verification Simulator is an automated backup validation platform designed to ensure that database backups are not only created but also verified for recoverability and integrity. In many organizations, backups are generated regularly as part of disaster recovery planning; however, these backups are often stored without being tested. As a result, organizations may discover corrupted or unusable backups only during critical recovery situations.

This project addresses that problem by automating the entire backup verification lifecycle. The system periodically creates SQLite database backups, restores them into an isolated sandbox environment, performs multiple validation checks, generates verification reports, stores execution metadata, and automatically raises GitHub issues when failures are detected.

The objective of the project is to provide confidence that backups can be successfully restored when needed, thereby improving disaster recovery readiness and reducing operational risk.

Problem Statement

Organizations create backups daily to protect critical business data. However, backup creation alone does not guarantee successful recovery. In many cases, backups are never validated after creation, leading to situations where corrupted, incomplete, or inaccessible backups remain undetected until an actual disaster occurs.

Manual backup verification is time-consuming, repetitive, and prone to human error. Therefore, there is a need for an automated system that continuously verifies backup reliability and provides actionable insights when failures occur.

Proposed Solution

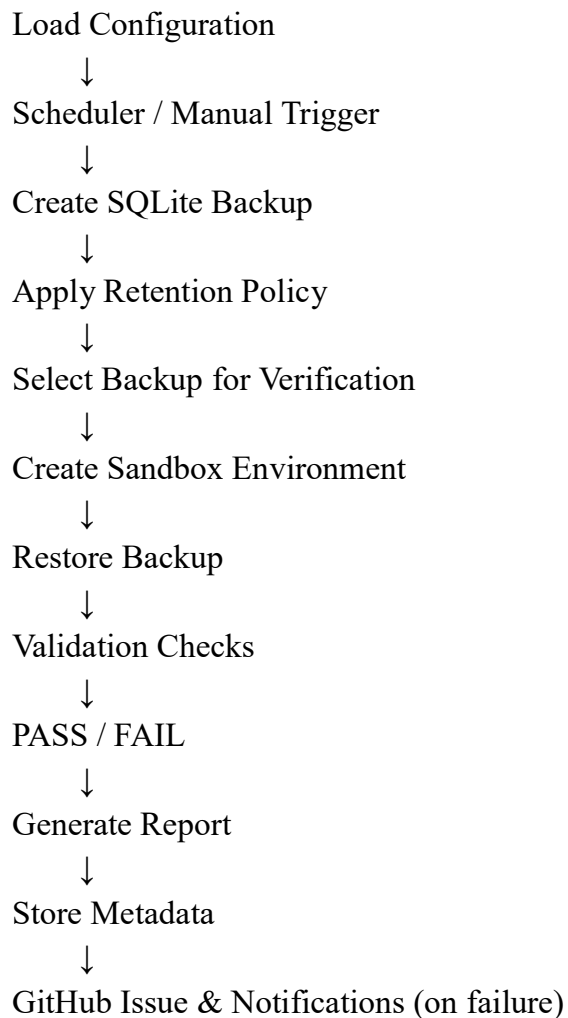
The Backup Verification Simulator automates the backup verification process through a workflow-driven architecture.

The system begins by loading user-defined configuration settings and triggering execution either manually or through a scheduler. A SQLite backup is created and stored according to a configurable retention policy. The workflow agent then selects a backup for verification and restores it into a sandbox environment where validation checks are performed safely without affecting production data.

The validation engine verifies database accessibility, table existence, row counts, SHA-256 checksum consistency, and SQLite integrity. Based on the validation outcome, the system generates either a success report or a failure report.

All execution details are stored in a metadata database for auditing and monitoring. If validation fails, a GitHub issue is automatically created and an email notification is sent to the configured recipients.

Workflow



Technology Stack

Backend

- Python

Web Framework

- FastAPI / Flask

Database

- SQLite

Automation

- Scheduler

Version Control

- Git & GitHub

Notifications

- Gmail Integration

Deployment

- Render

Monitoring

- Metadata Database & Logs

Expected Outcome

The project ensures that backups are continuously verified rather than simply stored. By automating restoration, validation, reporting, and issue tracking, the system improves backup reliability, minimizes recovery risks, reduces manual effort, and helps organizations maintain disaster recovery readiness.

GITHUB LINK: https://github.com/Vannilavan05/IM_INFI

RENDER LINK: <https://infi-scheduler.onrender.com/>